

SIMATS ENGINEERING

ASSESSMENT TOOL 7: Experiments

**COURSE NAME: Cloud Computing and Big
Data Analytics**

COURSE CODE: CSA1511

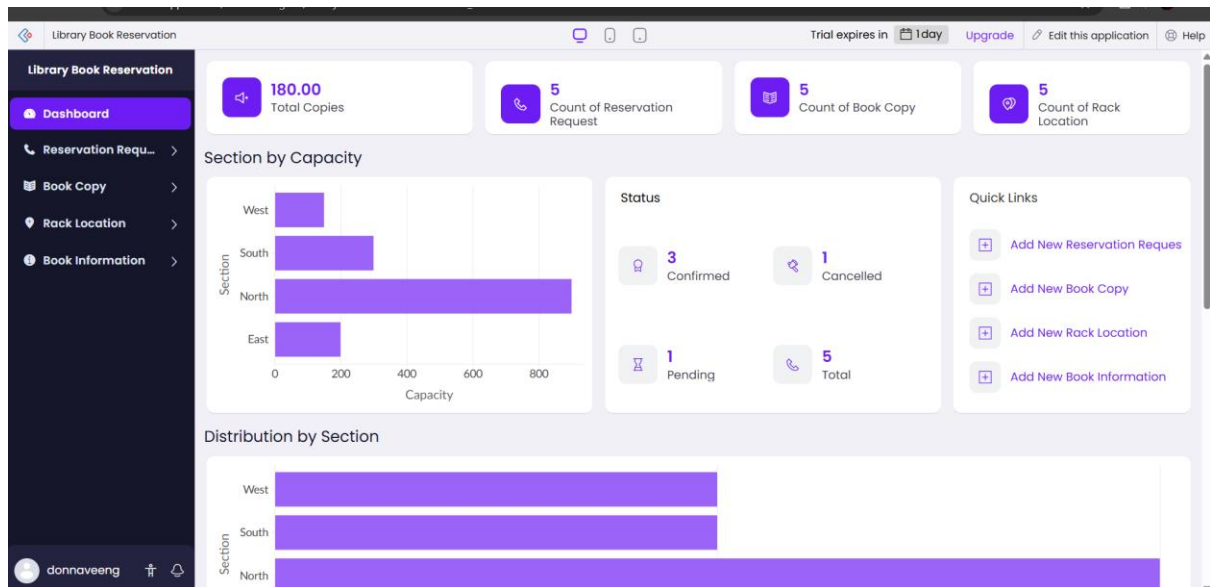
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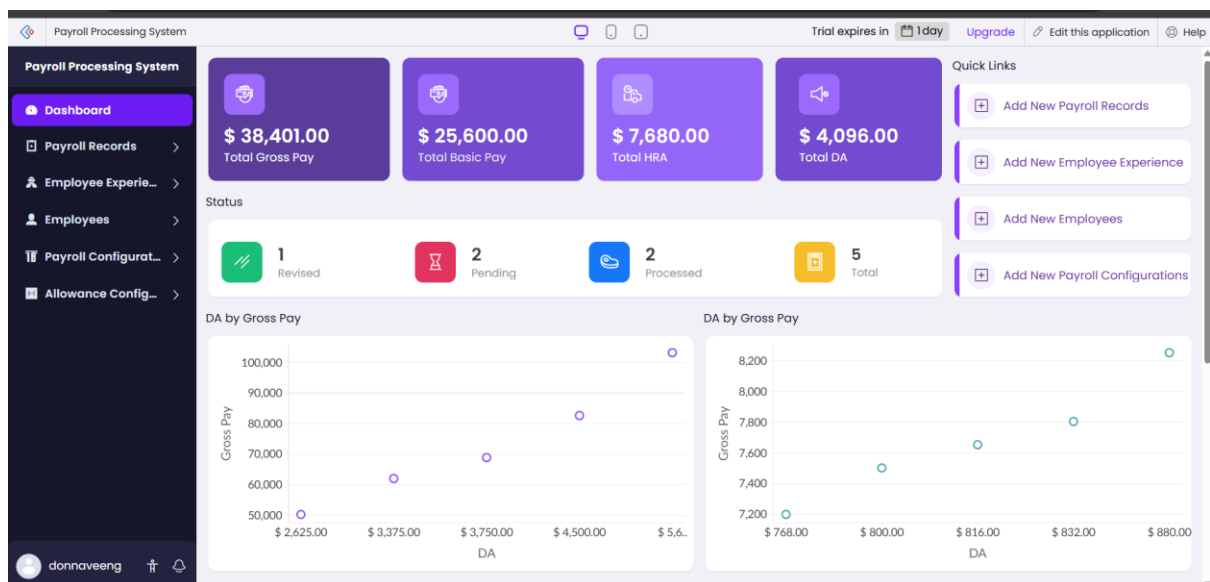
COURSE OUTCOME COVERED

**CO3: Implement cloud-based solutions using
cloud storage, web APIs, and platform services
for real-world**

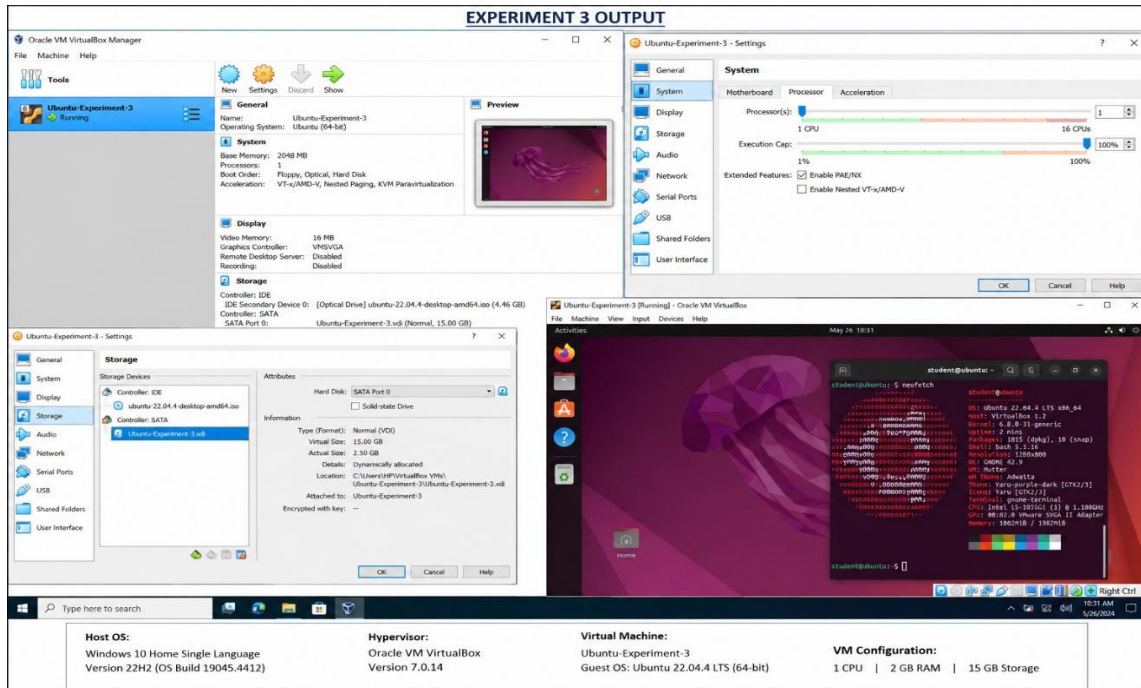
Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.



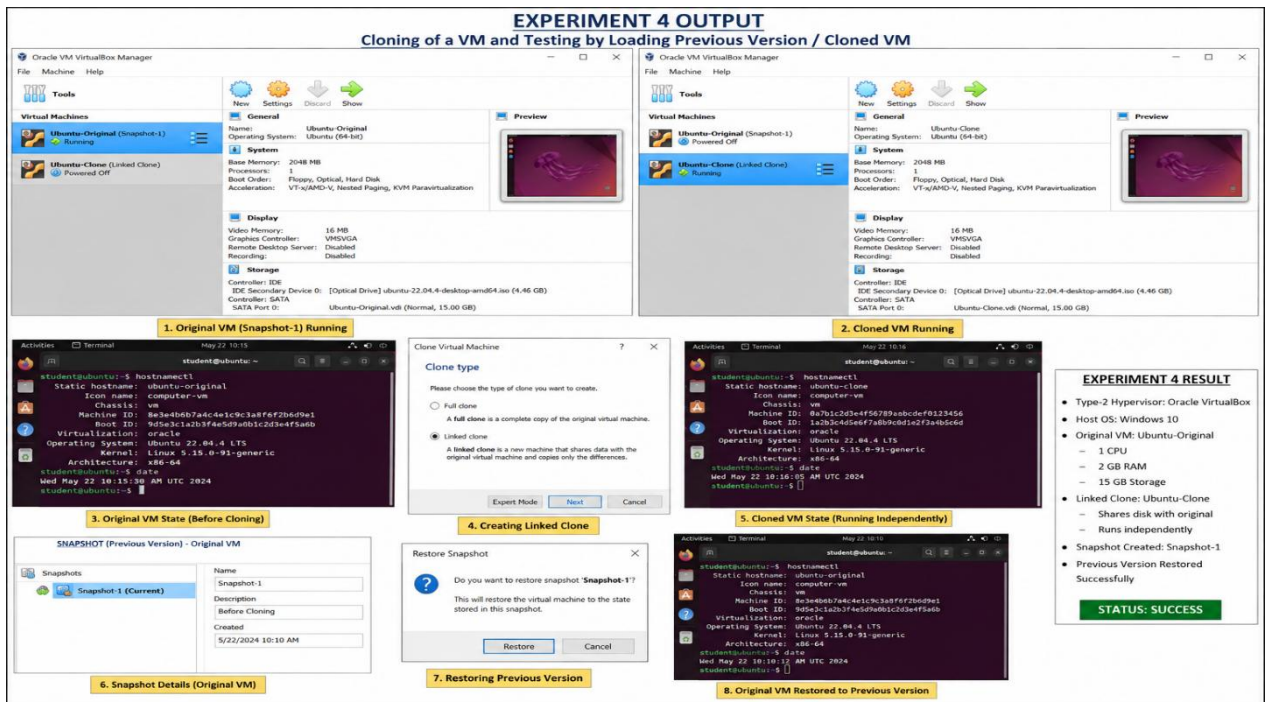
Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org, over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.



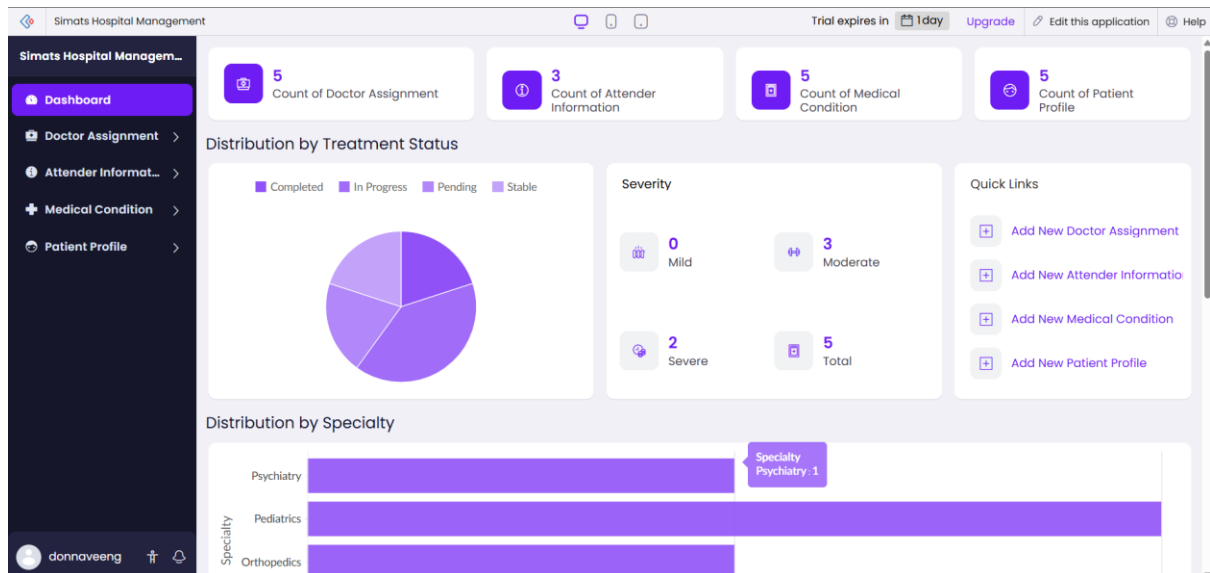
Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.



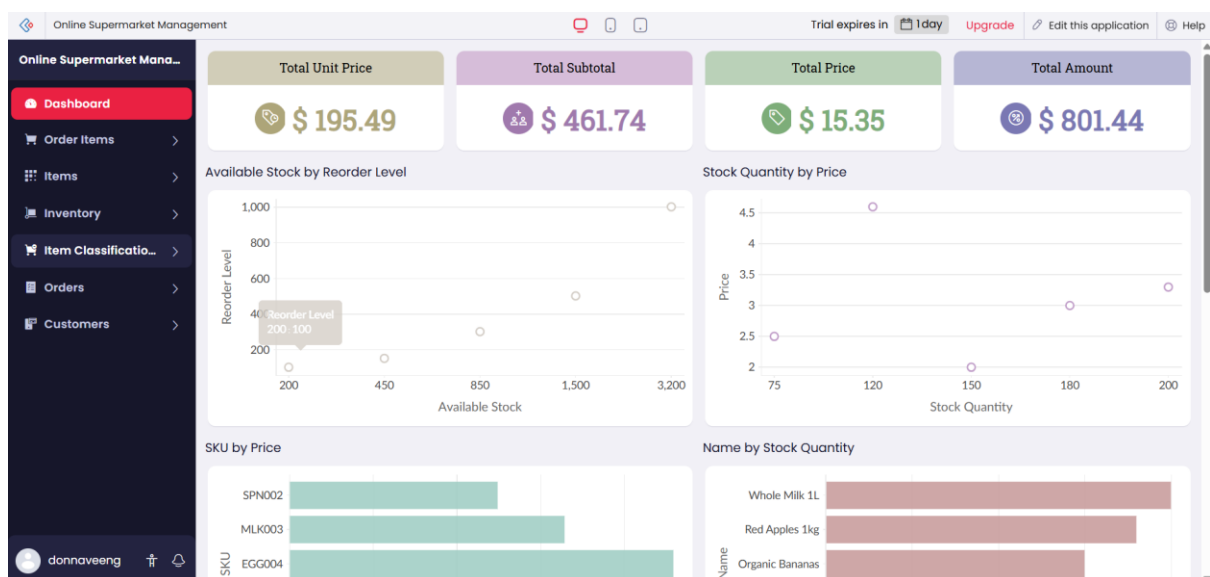
Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM



Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.



Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.



Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

EXPERIMENT 3 OUTPUT

Type-2 Hypervisor: Oracle VirtualBox | VM Configuration: 3 CPU, 10GB RAM, 10GB Storage

Host OS Details

OS: Windows 10 Home Single Language
Version: 22H2 (OS Build 19045.4412)

Hypervisor Details

Hypervisor: Oracle VirtualBox
Version: 7.0.14

Virtual Machine Details

Name: Ubuntu-Exp3
Guest OS: Ubuntu 22.04.4 LTS (64-bit)

VM Configuration

- CPU: 3
- RAM: 10 GB
- Storage: 10 GB (VDI)

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM

EXPERIMENT 4 OUTPUT

Snapshot of a VM and Test by Loading the Previous Version

Host OS Details

Windows 10 Home Single Language
Version: 22H2 (OS Build 19045.4412)

Hypervisor Details

Oracle VirtualBox
Version: 7.0.14

VM Details

Name: Ubuntu-Snapshot-VM
Guest OS: Ubuntu 22.04.4 LTS (64-bit)

VM Configuration

- CPU: 3
- RAM: 10 GB
- Storage: 10 GB (VDI)

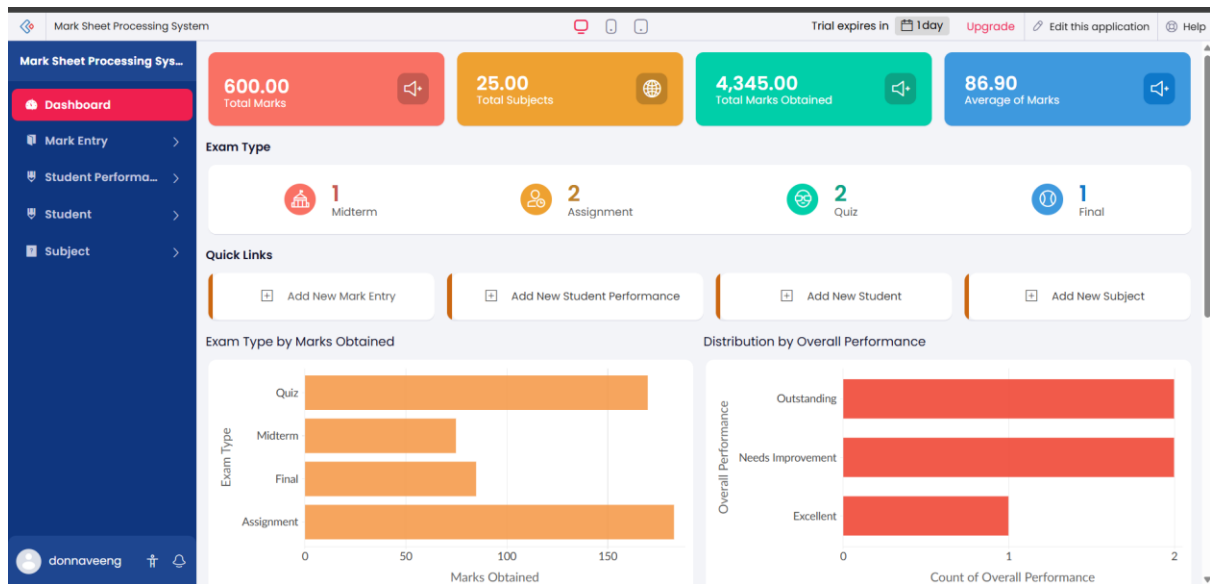
EXPERIMENT 4 RESULT

- Type-2 Hypervisor: Oracle VirtualBox
- Host OS: Windows 10
- Guest OS: Ubuntu 22.04.4 LTS
- VM Configuration:
 - CPU: 3
 - RAM: 10 GB
 - Storage: 10 GB
- Snapshot Name: Snapshot-1
- Description: Initial clean state
- Changes After Snapshot:
 - Created a file "testfile.txt"
- Restored Snapshot: Success
- Result: Previous state loaded successfully (file removed)

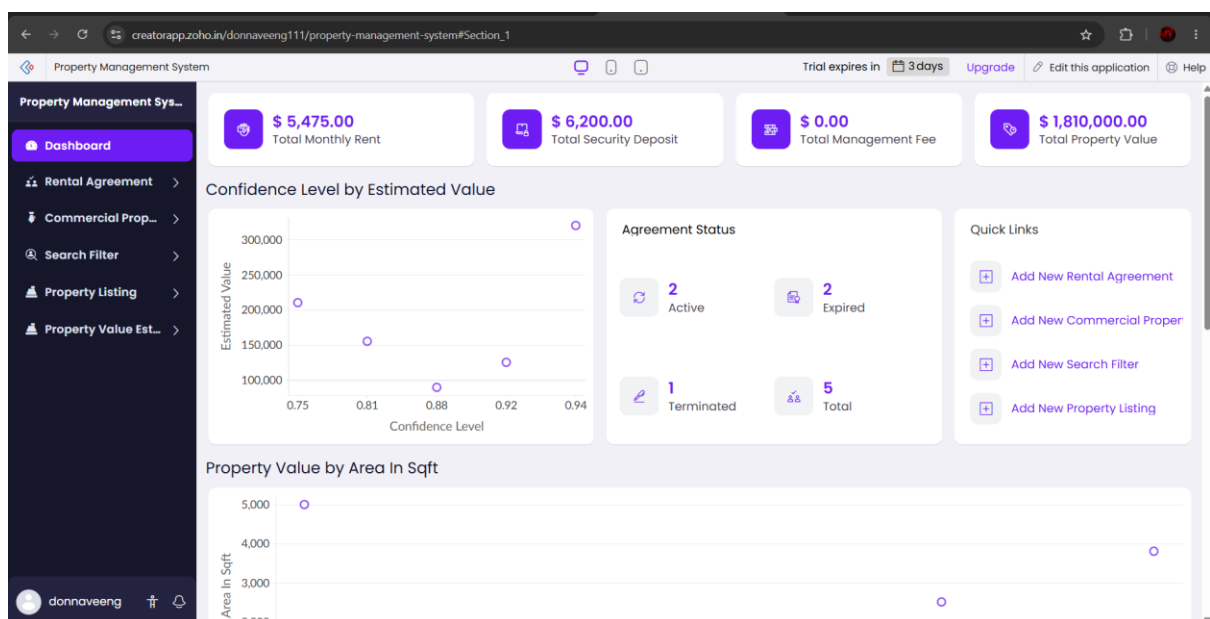
STATUS: SUCCESS

Assessment Tasks

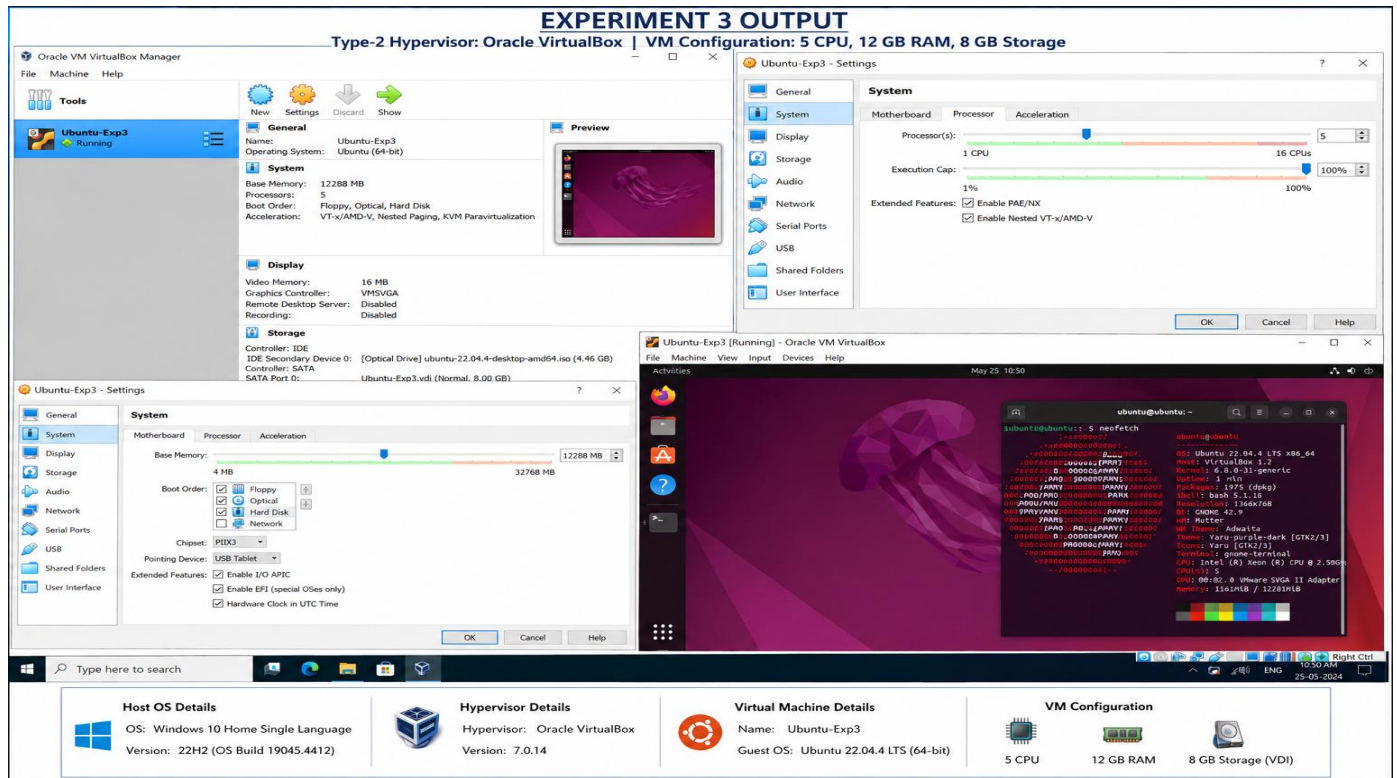
Experiment 1: Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the student, Reg no of the student, Subjects, Marks, Average, percentage, rank, overall performance etc.



Experiment 2: Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Buy, Rent, commercial, Rental Agreement, Rental Agreement, Area, place, Range, property value etc.



Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 5 CPU, 12 GB RAM and 8 GB storage disk using a Type 2 Virtualization Software.



Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

